

Quantum Information theory

Problem sheet -> Solution

Solution

2. a, $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} 3-i \\ 1+i \end{pmatrix}, \begin{pmatrix} i, 3 \end{pmatrix}, \begin{pmatrix} 0, 2 \end{pmatrix}$

then, I will put the column matrix in ket notation and the row matrix in bra notation

$$\left. \begin{array}{l} \text{Ket} - 1 \times 1 \\ \text{Bra} - 1 \times 1 \end{array} \right\} \text{Bra-Ket notation expression}$$

The first two are column matrix they should be put in ket notation and the last two are row matrix they put in bra notation

$$\textcircled{x} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left(\underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{|0\rangle} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{|1\rangle} \right) = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$\therefore \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle), \text{ which is already normalized}$$

$$\textcircled{x} \begin{pmatrix} 3-i \\ 1+i \end{pmatrix} \rightarrow \text{First it needed normalized (Norm)}$$

$$\begin{aligned} \text{Norm} &= \sqrt{|3-i|^2 + |1+i|^2} = \sqrt{|18-6i| + |2i|} = \sqrt{(\sqrt{3})^2 + (\sqrt{6})^2 + \sqrt{2}^2} \\ &= \sqrt{\sqrt{64+36} + \sqrt{4}} = \sqrt{\sqrt{100} + 14} = \sqrt{20 + 2} = \sqrt{12} \\ &= 2\sqrt{3} \end{aligned}$$

$$\begin{pmatrix} 3-i \\ 1+i \end{pmatrix} = \begin{pmatrix} 3-i \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1+i \end{pmatrix} = 3-i \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1+i \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

To put in ket & divided by norm

$$|\psi\rangle = \frac{(3-i) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + (1+i) \begin{pmatrix} 0 \\ 1 \end{pmatrix}}{2\sqrt{3}} = \frac{(3-i)|0\rangle + (1+i)|1\rangle}{2\sqrt{3}}$$

$$|\psi\rangle = \frac{(3-i)|0\rangle + (1+i)|1\rangle}{2\sqrt{3}} = \frac{1}{2\sqrt{3}} ((3-i)|0\rangle + (1+i)|1\rangle)$$

Now vectors - bra

$$(x) \quad (1 \ 3) = 1\langle 0| + 3\langle 1|$$

To normalize, compute norm

$$\text{Norm} = \sqrt{|1|^2 + |3|^2} = \sqrt{1+9} = \sqrt{10}$$

$$\langle \psi| = \frac{1}{\sqrt{10}} (1\langle 0| + 3\langle 1|) \checkmark$$

$$(x) \quad (0 \ 2) = 0\langle 0| + 2\langle 1| = 2\langle 1|$$

$$\text{norm} = \sqrt{(0)^2 + (2)^2} = \sqrt{4} = 2$$

$$\langle \psi| = \frac{2\langle 1|}{2} = \langle 1|$$

$$\langle \psi| = \langle 1| \checkmark$$

(b) ket: $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\langle 0| = \begin{pmatrix} 1 & 0 \end{pmatrix}$, $\langle 1| = \begin{pmatrix} 0 & 1 \end{pmatrix}$

An outer product like $|i\rangle\langle j|$

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad |1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Now put the Pauli matrix in bra-ket

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{|0\rangle\langle 1|} + \underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}_{|1\rangle\langle 0|} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \underbrace{|0\rangle\langle 1| + |1\rangle\langle 0|}_\checkmark$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ i & 0 \end{pmatrix} = -i \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{|0\rangle\langle 1|} + i \underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}_{|1\rangle\langle 0|}$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \underline{\underline{-i|0\rangle\langle 1| + i|1\rangle\langle 0|}}$$

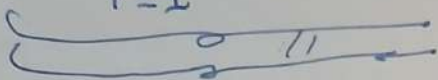
$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{|0\rangle\langle 0|} - \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{|1\rangle\langle 1|}$$

$$Z = \underline{\underline{|0\rangle\langle 0| - |1\rangle\langle 1|}}$$

2. A matrix A has eigenvalues λ_i with associated eigenstates $|\lambda_i\rangle$, where $i = 1, \dots, d$

a) Express the operator form of A in terms of its eigenvalues in bra-ket notation.

Solution

$$A = \sum_{i=1}^d a_i |a_i\rangle \langle a_i|$$


(b) Then,

Solution

The matrix D is diagonal in the computational basis hence we can write

$$D = \sum_{i=1}^d d_i |i\rangle \langle i| \text{ where } \{d_i\}_{i=1}^d \text{ are the singular values of}$$

A . we expand.

$$\begin{aligned} A &= U D U^\dagger \\ &= U \left(\sum_{i=1}^d d_i |i\rangle \langle i| \right) U^\dagger \\ &= \sum_{i=1}^d d_i U |i\rangle \langle i| U^\dagger \end{aligned}$$

Now let us define the vectors

$|u_i\rangle = U |i\rangle$ and $|v_i\rangle = U^\dagger |i\rangle$ Then we have the form

$$A = \sum_{i=1}^d d_i |u_i\rangle \langle v_i|$$


3. Consider a state $|u\rangle \in \mathbb{C}^d \otimes \mathbb{C}^{d'}$ with

$$|u\rangle = \sum_{i=1}^d \sum_{j=1}^{d'} V_{ij} |i\rangle \otimes |j\rangle$$

(a) what is the dimension of the Hilbert space given by $\mathbb{C}^d \otimes \mathbb{C}^{d'}$.

Solution

$$\boxed{\dim(C \otimes C') = d \times d'}$$

(b) How would one determine the probability amplitudes V_{ij} given the above expression

HPM, what is $(\langle i| \otimes \langle j|) (|i'\rangle \otimes |j'\rangle)$?

Solution

From the given above we have

$$V_{ij} = (\langle i| \otimes \langle j|) |V\rangle = \langle i, j | V \rangle$$

Because

$$(\langle i| \otimes \langle j|) |V\rangle = (\langle i| \otimes \langle j|) \sum_{i'=1}^d \sum_{j'=1}^{d'} V_{i'j'} (|i'\rangle \otimes |j'\rangle)$$

$$= \sum_{i'=1}^d \sum_{j'=1}^{d'} V_{i'j'} (\langle i| \otimes \langle j|) (|i'\rangle \otimes |j'\rangle)$$

$$= \sum_{i'=1}^d \sum_{j'=1}^{d'} V_{i'j'} (\underbrace{\langle i|i'\rangle}_{\delta_{ii'}} \otimes \underbrace{\langle j|j'\rangle}_{\delta_{jj'}})$$

$$= \sum_{i'=1}^d \sum_{j'=1}^{d'} V_{i'j'} \delta_{ii'} \delta_{jj'}$$

$$= V_{ij} \text{ where } i=i' \text{ and } j=j' \text{ } \left(\sum_{i'=1}^d \delta_{ii'} \right) \left(\sum_{j'=1}^{d'} \delta_{jj'} \right) = (1)(1)$$

Therefore

$$\underline{(\langle i| \otimes \langle j|) |V\rangle = V_{ij}}$$

4. The tensor product of two operators can be defined by

$$(A \otimes B)(|u\rangle \otimes |v\rangle) = (A|u\rangle) \otimes (B|v\rangle)$$

a) Show, $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$

Solution

we have that

$$\begin{aligned} (A \otimes B)(C \otimes D)(|u\rangle \otimes |v\rangle) &= (A \otimes B)(C|u\rangle \otimes D|v\rangle) \\ &= AC|u\rangle \otimes BD|v\rangle \\ &= (AC \otimes BD)(|u\rangle \otimes |v\rangle) \end{aligned}$$

$$(A \otimes B)(C \otimes D)(|u\rangle \otimes |v\rangle) = (AC \otimes BD)(|u\rangle \otimes |v\rangle)$$

~~$$(A \otimes B)(C \otimes D)(|u\rangle \otimes |v\rangle) = (AC \otimes BD)(|u\rangle \otimes |v\rangle)$$~~

For all $|u\rangle, |v\rangle$ given

$$(A \otimes B)(C \otimes D) = (AC \otimes BD)$$

(Spanning the tensor space product vectors span)

b) If $A = \sum_{i,i'=1}^d A_{ii'} |i\rangle \langle i'|$ and $B = \sum_{j,j'=1}^d B_{jj'} |j\rangle \langle j'|$ determine the bra-ket

Expansion of $A \otimes B$

Solution

We know that any matrix C can be expressed in an

orthonormal basis $|i\rangle \langle j|$ as

$$\sum_{i=1}^d \sum_{j=1}^d C_{ij} |i\rangle \langle j|$$

Therefore the action G is completely determined by knowing all terms

$$\langle i | \langle j |$$

in the case of $(A \otimes B)$

$$(\langle i | \otimes \langle j |) (A \otimes B) (|i'\rangle \otimes |j'\rangle) = (\langle i | \otimes \langle j |) (A|i'\rangle) \otimes (B|j'\rangle)$$

$$= (\langle i | \otimes \langle j |) \left(\sum_{i''=1}^d \sum_{j''=1}^d A_{i''i'} |i''\rangle \right) \otimes \left(\sum_{j''=1}^d \sum_{j'''=1}^d B_{j''j'} |j''\rangle \right)$$

$$= (\langle i | \otimes \langle j |) \left(\sum_{i''=1}^d A_{i''i'} |i''\rangle \right) \otimes \left(\sum_{j''=1}^d B_{j''j'} |j''\rangle \right) \underbrace{\sum_{i'''=1}^d \sum_{j'''=1}^d \langle i | i'' \rangle \langle j | j'' \rangle}_{1}$$

where, $\sum_{i''=1}^d \sum_{j''=1}^d \langle i | i'' \rangle \langle j | j'' \rangle = \delta_{i,i''} \delta_{j,j''}$
 where $i''=i'$ and $j''=j'$

hence, therefore,

$$= (\langle i | \otimes \langle j |) \left(\sum_{i''=1}^d A_{i''i'} |i''\rangle \right) \otimes \left(\sum_{j''=1}^d B_{j''j'} |j''\rangle \right)$$

$$= \underline{\underline{A_{i''i'} B_{j''j'}}}$$

Therefore $A \otimes B$ can be written as

$$A \otimes B = \sum_{i,j=1}^d \sum_{i',j'=1}^d A_{ii'} B_{jj'} (|i\rangle \otimes |j\rangle) (\langle i'| \otimes \langle j'|)$$

$$= \sum_{i,i'=1}^d \sum_{j,j'=1}^d A_{ii'} B_{jj'} (|i\rangle \langle i'|) \otimes (|j\rangle \langle j'|)$$

$$= \sum_{i,j=1}^d \sum_{i',j'=1}^d A_{ii'} B_{jj'} |i,j\rangle \langle i',j'|$$

c) write the following matrix tensor product, where x, y, z are defined in the first question

$$y \otimes z, z \otimes x$$

Solution

$$y \otimes z = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix}$$

$$y \otimes z = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix}$$

$$z \otimes x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & -i & 0 \end{pmatrix}$$

5 Are there states $|u\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^2$ that cannot be expressed as the tensor product $|u\rangle = |u_1\rangle \otimes |u_2\rangle$ with $|u_1\rangle, |u_2\rangle \in \mathbb{C}^2$? Give an example of such states?

Solution

Yes there is state that can't be expressed like $|u\rangle = |u_1\rangle \otimes |u_2\rangle$

$$|u\rangle = |u_1\rangle \otimes |u_2\rangle \rightarrow \text{separable state}$$

$$|u\rangle \neq |u_1\rangle \otimes |u_2\rangle \rightarrow \text{Entangled state / not separable}$$

The four Bell states are example of $|u\rangle \neq |u_1\rangle \otimes |u_2\rangle$

Example

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

There is no any measurement that make this state separable

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

Are all maximal entangled state

Problem 2: Fundamental of Quantum mechanics

1. Prove that the eigenvalue of a Hermitian operator are real

Solution

Let's say H is Hermitian operator

$$H = H^\dagger$$

where

$$H|\psi\rangle = \lambda|\psi\rangle \quad \dots \dots \dots \textcircled{x}$$

Multiply both side by $\langle\psi|$

$$\langle\psi|H|\psi\rangle = \langle\psi|\lambda|\psi\rangle$$

$$\langle\psi|H|\psi\rangle = \lambda\langle\psi|\psi\rangle$$

$$\langle\psi|H|\psi\rangle = \lambda\langle\psi|\psi\rangle \quad \dots \dots \dots \times \times$$

Take the complex conjugate of both sides

$$(\langle\psi|H|\psi\rangle)^* = (\lambda\langle\psi|\psi\rangle)^* \quad \left. \begin{array}{l} \text{where} \\ \lambda^*(\langle\psi|\psi\rangle)^* = \lambda^*\langle\psi|\psi\rangle \end{array} \right\}$$

$$\langle\psi|H^\dagger|\psi\rangle = \lambda^*\langle\psi|\psi\rangle$$

$$\lambda^*(\langle\psi|\psi\rangle)^* = \lambda^*\langle\psi|\psi\rangle$$

∴

$\langle\psi|\psi\rangle$ is already a real number

$H = H^\dagger$ therefore

$$\underbrace{\langle\psi|H|\psi\rangle}_{\lambda\langle\psi|\psi\rangle} = \underbrace{\langle\psi|H^\dagger|\psi\rangle}_{\lambda^*\langle\psi|\psi\rangle} \quad \left. \begin{array}{l} \text{Both expressions} \\ \text{are equal} \end{array} \right\}$$

Compare

$$\lambda\langle\psi|\psi\rangle = \lambda^*\langle\psi|\psi\rangle, \quad \text{since } \langle\psi|\psi\rangle > 0$$

$$\boxed{\lambda = \lambda^*} \rightarrow \lambda \text{ is equal to complex conjugate of } \lambda \text{ i.e. } \lambda$$

and only if when λ is real

Also show that the eigenvectors of a Hermitian operator with different eigenvalues are orthogonal to each other.

Solution

V and W are eigenvectors of two different eigenvalues
 λ and μ

Then

$$\begin{aligned} H|W\rangle &= \lambda|W\rangle \\ H|W\rangle &= \mu|W\rangle \end{aligned}$$

First

$$\langle V|H|W\rangle = \langle H^\dagger V|W\rangle = \langle H V|W\rangle \dots \times$$

Therefore

$$\langle V|H|W\rangle = \langle H^\dagger V|W\rangle = \langle H V|W\rangle$$

$$\underbrace{\langle V|H|W\rangle}_{\langle V|W\rangle} = \underbrace{\langle H V|W\rangle}_{\langle \lambda V|W\rangle}$$

$$\langle V|W\rangle = \langle \lambda V|W\rangle$$

$$\langle V|W\rangle = \lambda^* \langle V|W\rangle$$

$$\langle V|W\rangle - \lambda^* \langle V|W\rangle = 0$$

$$(\langle V|W\rangle - \lambda^* \langle V|W\rangle) = 0 \quad \text{since } \boxed{\lambda^* = \lambda} \text{ - from above}$$

$$(\langle V|W\rangle - \lambda \langle V|W\rangle) = 0$$

which means that either

$$u \cdot v = 0 \text{ or } |u| |v| > 0$$

where u and v are different values. Leaver this

$$|u| |v| = 0, \text{ verifying}$$

$$\frac{|u| |v| = 0 \Rightarrow u \perp v}{\text{Hence.}}$$

2. Given

$$S = \sum_i p_i |\psi_i\rangle \langle \psi_i|$$

Product rule

$$(AB)^{\dagger} = B^{\dagger} A^{\dagger} \quad \left. \begin{array}{l} p_i \geq 0, (p_i \in \mathbb{R}) \end{array} \right\}$$

a) required
 $S^{\dagger} = S$

Solution

$$S^{\dagger} = \left(\sum_i p_i |\psi_i\rangle \langle \psi_i| \right)^{\dagger} = \sum_i p_i (|\psi_i\rangle \langle \psi_i|)^{\dagger}, \quad p_i \in \mathbb{R}$$

$$= \sum_i p_i (|\psi_i\rangle)^{\dagger} (\langle \psi_i|)^{\dagger}, \text{ where product rule } (AB)^{\dagger} = B^{\dagger} A^{\dagger}$$

therefore

$$\left. \begin{array}{l} (|\psi_i\rangle)^{\dagger} = \langle \psi_i| \\ (\langle \psi_i|)^{\dagger} = |\psi_i\rangle \end{array} \right\}$$

$$S^{\dagger} = \underbrace{\sum_i p_i |\psi_i\rangle \langle \psi_i|}_S$$

$S^{\dagger} = S$ which density operator is Hermitian

b) Given

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

Required

$$\text{Tr}[\rho] = 1$$

Solution

$$\text{Tr}[\rho] = \text{Tr}\left[\sum_i p_i |\psi_i\rangle\langle\psi_i|\right] = \sum_i p_i \text{Tr}[\langle\psi_i|\psi_i\rangle]$$

from trace identity

$$\text{Tr}[|a\rangle\langle b|] = \langle b|a\rangle$$

Therefore

$$\text{Tr}[\rho] = \sum_i p_i \text{Tr}[\langle\psi_i|\psi_i\rangle]$$

$$= \sum_i p_i \langle\psi_i|\psi_i\rangle, \text{ where } |\psi_i\rangle \text{ is normalized state}$$

$$= \sum_i p_i (1), \text{ where } \sum_i p_i = 1$$

$$= 1 \times 1$$

$$\text{Tr}[\rho] = \underline{\underline{1}}, \text{ ~~therefore~~}$$

c) Given

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

Required

$$\rho \geq 0$$

Solution

Let's say any state $|\phi\rangle$

Compute $\langle\phi|\rho|\phi\rangle$

$$\langle\phi|\rho|\phi\rangle = \langle\phi|\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|\right)|\phi\rangle = \sum_i p_i \langle\phi|\psi_i\rangle\langle\psi_i|\phi\rangle$$

By - Inner product: modulus identity / projection identity

$$\boxed{\langle \phi | \psi \rangle \langle \psi | \phi \rangle = |\langle \psi | \phi \rangle|^2}$$

Projection identity

$\langle \phi | \rho | \phi \rangle = \sum_i p_i |\langle \psi_i | \phi \rangle|^2$, where, $|\langle \psi_i | \phi \rangle|^2 \geq 0$ is the norm which is non-negative

$$= \sum_i p_i |\langle \psi_i | \phi \rangle|^2 \geq 0$$

Therefore,

$$\boxed{\rho \geq 0}$$

d) Given

$$\rho^\dagger = \rho$$

$$I^\dagger = I$$

$$\text{Tr}(\rho) = 1$$

$$\text{Tr}(I) = \dim H (d)$$

Required

$$\frac{1}{\dim H} \leq \text{Tr}(\rho^2) \leq 1$$

Cauchy - Schwarz Inequality

$$(\text{Tr}[AB])^2 \leq \text{Tr}[A^\dagger A] \text{Tr}[B^\dagger B]$$

Solution

$\text{Tr}(\rho^2) = 1$ iff ρ is pure state

Lower bound $\text{Tr}(\rho^2) \geq \frac{1}{d}$

Apply Cauchy - Schwarz Inequality

$$\text{with } \left. \begin{array}{l} A = \rho \\ B = I \end{array} \right\}$$

$$(\text{Tr}[\rho I])^2 \leq \text{Tr}[\rho^\dagger \rho] \text{Tr}[I^\dagger I] \quad \text{where } \left. \begin{array}{l} \rho I = \rho \\ I^\dagger = I \\ \rho^\dagger = \rho \end{array} \right\}$$

$$(\text{Tr}(\rho))^2 \leq \text{Tr}(\rho^2) \text{Tr}(I)$$

$$(\text{Tr}(\rho))^2 \leq \text{Tr}(\rho^2) \text{Tr}(\mathbb{I}) \quad , \quad \text{where } \text{Tr}(\rho) = 1 \quad \left. \begin{array}{l} \text{Tr}(\mathbb{I}) = \dim H \end{array} \right\}$$

$$(1)^2 \leq \text{Tr}(\rho^2) \dim H$$

$$\frac{1}{\dim H} \leq \frac{\text{Tr}(\rho^2)}{\dim H}$$

$$\frac{1}{\dim H} \leq \text{Tr}(\rho^2)$$



upper bound $\text{Tr}(\rho^2) \leq 1$ and $\text{Tr}(\rho^2) \geq 2$

Given

ρ was eigenvector & $\lambda_i \geq 0$ with

$$\left. \begin{array}{l} \lambda_i \geq 0 \\ \sum \lambda_i = 1 \end{array} \right\}$$

Required

$$\text{Tr}(\rho^2) = \text{Tr}(\rho) = 1, \quad \text{Tr}(\rho^2) = 1$$

Solution

For pure state $\rho = |\psi\rangle\langle\psi|$, we have that

$$\text{Tr}(\rho^2) = \text{Tr}(|\psi\rangle\langle\psi| |\psi\rangle\langle\psi|) = \text{Tr}(|\psi\rangle\langle\psi|)$$

$$\text{Tr}(|\psi\rangle\langle\psi|)$$

$$= \text{Tr}(|\psi\rangle\langle\psi|), \quad \text{where } \text{Tr}(|a\rangle\langle b|) = \langle b|a\rangle$$

$$\text{Tr}(\rho^2) = \langle\psi|\psi\rangle$$

$$\text{Tr}(\rho^2) = 1$$



Upper bound $\text{Tr}[P^2] \leq 1$

Required

$\text{Tr}[P^2] \leq 1$, show that

Solution

from the trivial identity

$$\left(\sum_i \lambda_i\right)^2 = \sum_i \lambda_i^2 + 2 \sum_{i < j} \lambda_i \lambda_j$$

$$\left. \begin{array}{l} \text{where } \lambda_i \geq 0 \\ 2 \sum_{i < j} \lambda_i \lambda_j \geq 0 \end{array} \right\}$$

Therefore

$$\boxed{\sum_i \lambda_i^2 \leq \left(\sum_i \lambda_i\right)^2}, \text{ where } \sum_i \lambda_i = 1$$

$$\text{Tr}[P^2] = \sum_i \lambda_i^2$$

$$\text{Tr}[P^2] \leq \left(\underbrace{\sum_i \lambda_i}_1\right)^2$$

$$\text{Tr}[P^2] \leq 1$$

Therefore,

lower bound $\leq \text{Tr}[P^2] \leq$ upper bound

$$\boxed{\frac{1}{\dim \mathcal{H}} \leq \text{Tr}[P^2] \leq 1} =$$

At $\frac{1}{\dim \mathcal{H}} \rightarrow$ Maximally mixed state
 $1 \rightarrow$ pure state

c) Given an example of a state the attains the lower bound
 i.e. a state that satisfies $\text{Tr}(\rho^2) = \frac{1}{\dim \mathcal{H}}$

$$\text{Tr}(\rho^2) = \frac{1}{\dim \mathcal{H}}$$

③ which is maximally mixed state, it represents a mixture of all possible pure state of the system

↳ least pure state

↳ $\text{Tr}(\rho^2) \rightarrow$ purity, which determines

"How the state is to being pure?"

Higher values has higher purity

Lower values has higher degree of mixedness

$$\text{Tr}(\rho^2) = \frac{1}{\dim \mathcal{H}}$$

When we represented on the Bloch sphere, it found at center of Bloch sphere

Example ~~$\rho = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$~~

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

~~$\rho = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$~~

3. Starting from the Schrodinger equation for pure state $|\psi\rangle$

$$S = |\psi\rangle\langle\psi|$$

Show that, $\frac{dS}{dt} = -\frac{i}{\hbar} [H, S]$

Given

$$S = |\psi\rangle\langle\psi|$$

Required

$$\frac{dS}{dt} = -\frac{i}{\hbar} [H, S] \rightarrow \text{Show that?}$$

Requires

$$\frac{dS}{dt} = \frac{d(|\psi\rangle\langle\psi|)}{dt} = \frac{d(|\psi\rangle)}{dt} \langle\psi| + |\psi\rangle \frac{d(\langle\psi|)}{dt} \quad \dots (1)$$

From Schrodinger equation

$$\left. \begin{aligned} H|\psi\rangle &= i\hbar \frac{d|\psi\rangle}{dt} \Rightarrow \frac{d(|\psi\rangle)}{dt} = -\frac{i}{\hbar} (H|\psi\rangle) \\ \frac{d(\langle\psi|)}{dt} &= \frac{i}{\hbar} (\langle\psi|H) \end{aligned} \right\} \dots (2)$$

Substituted eq. 2 into eq. 1

$$\begin{aligned} \frac{dS}{dt} &= -\frac{i}{\hbar} (H|\psi\rangle)\langle\psi| + |\psi\rangle \frac{i}{\hbar} (\langle\psi|H) \\ &= -\frac{i}{\hbar} \underbrace{H|\psi\rangle\langle\psi|}_S + \frac{i}{\hbar} \underbrace{|\psi\rangle\langle\psi|H}_S, \text{ where } |\psi\rangle\langle\psi| = S \end{aligned}$$

$$\frac{dS}{dt} = -\frac{i}{\hbar} HS + \frac{i}{\hbar} SH = -\frac{i}{\hbar} (HS - SH) = -\frac{i}{\hbar} [H, S]$$

$$\frac{dS}{dt} = -\frac{i}{\hbar} [H, S] \quad \text{where, } [H, S] = [HS - SH]$$